

Richard B. Anderson

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Education

The University of Texas at El Paso (UTEP)

Bachelor of Science in Physics

Bachelor of Science in Mathematics

Anticipated Graduation : May 2026

Overall GPA: 3.79/4.00

Dean's List : Fall 2023 - Spring 2025

Research Experience

SULI Intern

Brookhaven National Laboratory - Radiation Detection
and Emerging Safeguards Technologies Group

June 2025 - August 2025

- Conducted discrete-element simulations in LIGGGHTS to generate and analyze realistic pebble-bed geometries, optimizing core fill and packing configurations for transient flow studies.
- Performed Monte Carlo neutron and γ -ray transport calculations in OpenMC to produce spectral signatures of individual fuel pebbles at multiple burnup stages, varying parameters such as initial enrichment, fuel composition, and irradiation history.
- Developed Python scripts to automate parameter sweeps, batch-process simulation outputs, and assemble a comprehensive dataset of neutron/ γ spectra for downstream machine-learning model training.
- Drafted a preliminary scientific report summarizing methodology, key findings, and recommendations for real-time, non-destructive burnup estimation via deep-learning regressors.

Undergraduate Research Assistant

UTEP - Astrophysics Theory Group

August 2022 - Present

- Conducting astrophysical research under Dr. Kedron Silsbee, studying the capture rate of planetesimals by collapsing proto-stars.
- Utilized Slurm *High-Performance Computing* cluster to run simulations via Python's *Rebound* package, and analyze subsequent data to come to a physically significant conclusion.
- Utilized knowledge of classical mechanics and mathematics to create analytical models of celestial bodies, as well as statistical predictions of particle motion.
- Presented preliminary findings at COURI Symposium poster sessions.

Undergraduate Research Intern - The University of Texas at Arlington

The University of Texas at Arlington Neutrino
and Rare Event Searches (NuRES) Group

May 2024 - December 2024

- Conducted research on neutrinoless double beta decay using NEXT experimental data.
- Analyzed S1 and S2 signals to improve detector sensitivity.
- Simulated particle events using the Geant-4 Nexus framework to study alpha-induced background radiation.
- Automated waveform analysis across large datasets, extracting key parameters like signal decay constants and peak widths.
- Developed visualizations for detector event data and contributed to the understanding of xenon gas ionization patterns.
- Applied effective field theory techniques to explore beyond Standard Model physics in double beta decay.
- Used nuDoBE Python tool to calculate decay rates and electron kinematics under different Wilson coefficients.

Undergraduate Research Assistant - UTEP Aerospace Center

UTEP Aerospace Center - Missile Systems Innovation Team

September 2023 - March 2024

- Effectively employed *Python* machine learning and computer vision packages, such as *OpenCV* and *TensorFlow*, to implement neural networks for object detection and tracking.
- Utilized drone hardware and *ArduPilot* software for object tracking.
- Experience collaborating with a multidisciplinary team dedicated to utilizing digital engineering techniques to enhance and innovate missile systems, with a focus on achieving *cost-effective solutions*.
- Applied numerical and analytical techniques to calculate ballistic trajectories and optimize missile sizing and design parameters.

Teaching Experience

Academic Tutor

El Paso Community College - Student Success

July 2022 - July 2023

- Assisted students of El Paso Community College with mathematics coursework in courses ranging from remedial mathematics courses to differential equations.
 - Coordinated with other tutoring team members during high-traffic hours to efficiently and effectively address the high volume of students.
 - Utilized both in person tutoring sessions, and navigated online learning interface to deliver online tutoring option for students.
 - Proficiently utilized Microsoft Suite programs for seamless communication within the tutoring team and to meticulously track student data.
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Awards & Activities

Terry Scholarship

The Terry Foundation

Fall 2023 – Spring 2027

- Awarded a prestigious academic scholarship of \$25,000 per academic year, based on high academic achievement and leadership potential.
- Engage in up to 15 hours per semester of community service activities in the El Paso community.

COURI SURPASS Program

The University of Texas at El Paso

June 2023 – August 2023

- Participated in a Summer Research Experience for Undergraduates with a financial award of up to \$2,000 for the summer.
- Attended academic research workshops and professional development training relevant to scientific research.
- Conducted research under Dr. Kedron Silsbee, contributing to research in planet formation.

Nuclear Physics Summer School for Underrepresented Students (NuPEERS)

Dillard University, New Orleans, LA

June 2023

- Participated in a summer school program focused on nuclear physics, supported by the Department of Energy's Office of Nuclear Physics.
 - Attended lectures and hands-on workshops covering advanced topics in nuclear physics, including neutrino physics, machine learning, and surveys of modern nuclear physics research.
 - Collaborated with peers from Minority Serving Institutions (MSIs) to solve challenging problems and explore innovative research in nuclear science.
 - Received a financial award of \$500 for successful completion of the program.
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Leadership Experience

UTEP Astronomy Club - Secretary

- Founding member and officer of the UTEP astronomy club.
 - Playing a key role in organizing and hosting observational astronomy events, enhancing public engagement with astronomy through educational presentations and night sky viewings.
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Technical Skills

C | C++ | Python | Fortran | Bash | Mathematica | Git | SLURM HPC Clusters | HTML | LaTeX | Liggghts | StarCCM